

GRAZ THALERHOF, Austria

WMO: 112400

Lat: 46.9806N

Lon: 15.4400E

Elev: 1191

StdP: 14.07

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	16.0	6.6	8.1	19.5	10.6	9.9	22.5	20.5	34.2	17.2	36.5	1.8	170	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	88.2	69.1	85.4	68.5	82.4	67.1	71.6	82.5	70.1	81.1	68.7	78.9	7.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.2	108.4	75.1	66.6	102.3	73.4	65.5	98.5	72.3	36.1	82.6	34.8	80.9	33.6	78.8	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	13.8	11.1	5.4	93.1	5.9	3.8	1.2	95.8	-2.3	98.1	-5.6	100.2	-9.9	103.0
			5.1	74.4	5.9	1.7	0.9	75.6	-2.5	76.6	-5.8	77.6	-10.0	78.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	30.6	34.9	42.7	51.7	60.0	66.7	69.5	68.4	60.1	51.4	41.7	32.0	(6)
	DBStd	15.16	7.02	7.79	7.37	6.59	5.97	5.98	5.25	5.22	5.57	6.79	7.12	6.66	(7)
	HDD50	2212	603	423	240	58	3	0	0	0	2	66	259	558	(8)
	HDD65	5579	1068	842	690	398	177	52	19	25	165	422	698	1023	(9)
	CDD50	2540	0	1	14	110	313	500	606	571	304	110	11	0	(10)
	CDD65	434	0	0	0	0	22	102	160	132	18	0	0	0	(11)
	CDH74	4060	0	0	0	30	271	945	1478	1179	147	10	0	0	(12)
	CDH80	1174	0	0	0	1	47	274	467	370	15	0	0	0	(13)
(14) Wind	WSAvg	3.7	2.9	3.3	4.4	4.8	4.8	4.5	4.3	3.7	3.5	3.0	3.0	2.7	(14)
(15) Precipitation	PrecAvg	34.9	1.0	1.4	1.7	2.2	3.5	4.8	5.1	4.9	3.6	2.6	2.3	1.5	(15)
	PrecMax	56.8	2.6	4.4	5.0	5.2	8.6	10.6	10.9	9.3	7.4	8.5	5.1	4.6	(16)
	PrecMin	22.7	0.0	0.0	0.0	0.5	1.3	0.9	1.3	1.9	0.6	0.0	0.0	0.0	(17)
	PrecStd	6.4	0.7	1.1	1.1	1.2	1.7	1.9	2.4	1.8	1.7	1.9	1.3	1.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	62.6	69.0	77.5	84.9	91.0	92.8	91.8	82.0	75.4	65.8	53.3	(19)
		MCWB	44.3	49.1	52.2	59.0	64.7	69.1	70.9	70.1	67.0	61.7	55.5	47.1	(20)
	2%	DB	48.3	55.7	64.8	73.3	80.4	86.4	88.1	87.7	77.9	70.1	60.0	47.6	(21)
		MCWB	42.2	45.4	50.5	56.1	63.2	69.0	69.4	69.0	64.7	59.4	53.0	43.1	(22)
	5%	DB	44.2	51.5	60.6	69.7	76.9	82.8	84.8	84.2	75.0	66.5	55.4	44.1	(23)
		MCWB	39.5	43.4	48.8	55.0	61.6	67.4	68.4	68.5	63.2	57.9	50.8	41.0	(24)
	10%	DB	40.6	46.8	56.4	66.0	73.3	79.7	82.1	80.7	71.5	62.6	51.9	41.0	(25)
		MCWB	37.7	40.4	46.8	53.1	59.9	65.7	67.2	66.9	61.2	56.1	48.4	38.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.6	49.5	53.8	60.8	67.1	72.6	73.8	73.3	68.7	63.4	56.8	48.6	(27)
		MCDB	51.4	60.7	65.1	73.8	78.6	84.6	85.6	84.5	77.8	71.0	62.9	52.6	(28)
	2%	WB	43.0	46.4	51.5	58.3	65.0	70.4	71.9	71.1	66.5	61.3	54.1	44.2	(29)
		MCDB	46.7	54.1	61.6	69.2	76.0	81.8	82.5	82.3	74.9	67.7	58.4	46.2	(30)
	5%	WB	40.3	44.0	49.7	56.2	63.1	68.9	70.2	69.7	64.8	59.3	51.6	41.2	(31)
		MCDB	43.2	49.9	58.8	66.0	73.8	79.7	81.0	80.6	72.2	64.4	55.2	42.8	(32)
	10%	WB	37.6	41.4	47.8	54.2	61.3	67.3	68.6	68.1	63.1	57.3	49.2	39.1	(33)
		MCDB	39.8	45.7	55.2	63.5	70.7	77.0	78.8	77.7	69.2	62.0	51.6	40.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	17.1	19.8	20.8	20.3	20.0	20.4	20.1	18.6	17.6	12.9	12.0	(35)
		MCDBR	20.3	25.4	28.2	28.4	26.4	25.3	25.0	26.0	23.2	23.5	20.5	17.9	(36)
	5% WB	MCWBR	13.8	15.3	15.0	12.9	10.7	9.5	9.0	9.5	11.0	13.1	13.0	13.2	(37)
		MCWBR	18.6	23.4	25.7	24.7	23.6	23.3	22.8	23.3	20.8	20.6	17.8	15.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.304	0.331	0.370	0.416	0.418	0.424	0.426	0.426	0.397	0.370	0.337	0.304	(40)
		taud	2.433	2.374	2.304	2.201	2.245	2.268	2.261	2.273	2.327	2.380	2.430	2.472	(41)
	Ebn at Noon		249	264	269	266	269	267	265	259	257	246	233	236	(42)
		Edn at Noon	22	28	35	43	42	42	41	39	34	28	22	19	(43)
(44) All-Sky Solar Radiation	RadAvg		393	662	1010	1326	1577	1745	1745	1546	1110	747	414	325	(44)
	RadStd		49	91	122	145	164	125	137	172	143	78	62	48	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (97 neighbors)	+0.94	N/A	N/A	+1.57	+0.86	+0.72	-184	-252	+162	+103

Nomenclature: See separate page